

BROOKS, MI, USA

WMO: 725415

Lat: 42.251N

Lon: 84.956W

Elev: 287

StdP: 97.93

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 54821

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.0	-15.0	-22.3	0.5	-17.1	-18.9	0.7	-13.5	10.4	-3.4	9.0	-3.4	1.6	220	0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	32.2	22.8	30.7	22.1	28.9	21.4	25.0	29.0	23.9	28.0	23.0	27.1	3.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	19.3	27.3	22.7	18.0	26.1	22.0	17.3	25.5	77.8	28.9	73.1	28.1	69.7	26.9	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	7.9	7.0	-23.4	34.0	3.7	1.6	-26.1	35.2	-28.2	36.1	-30.3	37.0	-33.0	38.1		
(5)			-23.6	26.6	3.5	1.3	-26.1	27.6	-28.2	28.3	-30.2	29.1	-32.7	30.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	-4.4	-3.4	2.3	9.0	15.2	20.2	22.3	21.2	17.7	10.8	4.8	-1.3		
	DBStd	10.67	6.30	6.19	6.34	5.29	4.85	3.73	3.09	2.98	4.29	4.87	5.28	5.56		
	HDD10.0	1740	446	376	253	80	10	0	0	0	1	49	171	353		
	HDD18.3	3568	704	608	499	283	122	22	4	8	63	237	406	610		
	CDD10.0	1596	1	1	13	50	172	306	381	348	232	75	16	2		
	CDD18.3	381	0	0	1	3	26	78	127	98	43	5	0	0		
	CDH23.3	3341	0	0	7	32	238	686	1179	781	376	41	1	0		
	CDH26.7	987	0	0	1	3	55	214	396	206	107	6	0	0		
(14) Wind	WSAvg	2.7	3.3	3.2	3.0	3.3	2.7	2.2	1.9	1.7	1.9	2.6	3.0	3.2		
(15) Precipitation	PrecAvg	920	60	46	53	74	101	103	93	93	87	82	69	55		
	PrecMax	1077	111	99	101	158	198	162	182	210	272	185	121	94		
	PrecMin	728	16	13	14	8	42	37	21	27	31	9	29	24		
	PrecStd	96	31	26	23	32	49	36	39	42	54	43	29	19		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.6	15.9	24.0	27.1	30.8	32.8	34.1	32.5	32.2	27.3	21.3	15.3		
		MCWB	12.2	12.1	17.0	16.5	19.7	22.2	23.8	23.7	21.8	19.2	14.6	13.4		
	2%	DB	9.7	12.2	18.8	23.7	28.0	31.2	32.2	30.9	29.9	24.0	17.7	12.1		
		MCWB	7.8	9.6	14.0	15.2	19.1	21.6	23.6	22.6	21.7	17.4	12.9	10.5		
	5%	DB	6.6	7.8	15.3	21.3	26.3	29.1	30.8	28.8	27.5	21.7	15.9	8.6		
		MCWB	4.6	5.4	11.3	13.6	18.3	21.0	22.6	21.9	20.1	16.3	11.7	6.9		
	10%	DB	3.7	4.7	12.3	18.6	23.8	27.4	28.8	27.5	25.5	19.0	13.0	5.9		
		MCWB	2.5	2.5	8.7	12.4	17.4	20.1	21.7	21.1	19.3	15.1	9.5	4.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.6	13.4	17.1	19.2	22.6	24.6	27.1	25.8	24.6	21.4	16.4	13.7		
		MCDB	13.3	14.4	23.2	24.3	26.5	29.3	30.0	30.7	29.0	25.5	19.3	15.2		
	2%	WB	8.1	9.5	14.5	17.1	21.2	23.5	25.5	24.5	23.3	19.4	14.2	10.2		
		MCDB	8.8	11.5	17.8	20.9	25.5	28.2	29.2	28.1	27.5	22.2	16.7	11.6		
	5%	WB	4.9	5.7	12.0	15.1	19.9	22.6	24.2	23.4	21.9	17.4	12.4	7.0		
		MCDB	6.4	7.9	14.8	19.2	24.3	27.2	28.4	26.9	25.1	19.9	15.1	8.0		
	10%	WB	2.6	2.8	8.6	13.0	18.7	21.6	23.2	22.4	20.5	15.6	10.0	4.2		
		MCDB	3.7	4.6	11.4	17.6	22.5	25.7	27.3	25.8	23.8	18.8	12.8	5.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	8.4	9.8	11.6	11.3	10.9	11.5	11.0	11.6	10.3	8.5	6.6		
		MCDBR	9.1	10.8	13.9	14.6	13.3	12.5	12.8	12.3	13.1	13.5	11.5	9.6		
	5% WB	MCWBR	7.9	8.7	9.4	8.1	6.9	5.4	5.6	5.8	6.2	7.5	8.0	8.4		
		MCWBR	8.5	10.3	12.4	12.5	11.3	10.3	10.6	9.9	10.6	10.8	10.5	9.0		
(40) Clear-Sky Solar Irradiance	taub		0.298	0.358	0.348	0.391	0.424	0.443	0.430	0.434	0.388	0.349	0.317	0.302		
	taud		2.369	2.148	2.340	2.245	2.197	2.185	2.228	2.211	2.322	2.427	2.465	2.402		
	Ebn at Noon		844	825	895	877	853	835	842	825	841	837	818	805		
	Edh at Noon		81	119	111	132	142	145	137	135	113	90	74	73		
(44) All-Sky Solar Radiation	RadAvg		1.43	2.30	3.54	4.61	5.35	6.04	6.13	5.28	4.27	2.74	1.71	1.17		
	RadStd		0.14	0.32	0.35	0.43	0.51	0.39	0.29	0.26	0.39	0.27	0.23	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page